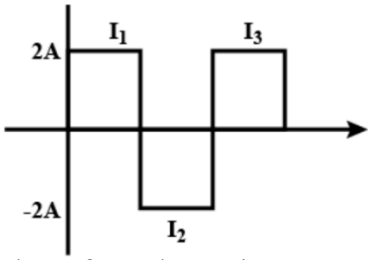
Chpater3 Electricity & Magnetism

Lesson3 AC & Transformer Systems

1. The instantaneous current and voltage of an ac circuit are given by: $i = 10 \sin 314t$ A and $v = 50 \sin 314t$ V. What is the power dissipation in the circuit?
2. An AC circuit has a frequency of 50 Hz and an RMS current of 5 A. If the current is initially zero and begins to increase in the positive direction, calculate the instantaneous current after $(1/300)$ seconds.
3. The electric mains in a house are marked $220\text{V}, 50\text{Hz}$. Write down the equation for instantaneous voltage.
4. Calculate the rms value of the alternating current shown in figure.



5. Define peak value and root mean value of an alternating current. Derive an expression for the root mean square value of alternating current.
6. **(a)** The peak voltage of an ac supply is 300 V. What is the rms voltage?

   (b) The rms value of current in an ac circuit is 10 A. What is the peak current?
7. An alternating voltage given by $\text{V} = 140 \sin 314t$ is connected across a pure resistor of $50\ \Omega$. Find **(i)** the frequency of the source **(ii)** the rms current through the resistor.
8. The peak value of an alternating voltage applied to a $50\ \Omega$ resistance is 10 V. Find the rms current, if the voltage frequency is 100 Hz, write the equation for the instantaneous current.
9. A $100\ \Omega$ resistor is connected to a 220 V, 50 Hz ac supply. **(a)** What is the rms value of current in the circuit? **(b)** What is the net power consumed over a full cycle?
10. A resistance of $40\ \Omega$ is connected to an ac source of 220 V, 50 Hz. Find **(i)** the rms current **(ii)** the maximum instantaneous current in the resistor and **(iii)** the time taken by the current to change from its maximum value to the rms value.
11. A step up transformer with 50 primary turns and 250 secondary turns is used to generate a current of 2.5 A at a voltage of 10 V. Find

    a) the turns ratio.
    b) the primary voltage.
    c) the primary current.
    d) the average power delivered to the secondary side.
    e) the average power of the primary side (by logic and by calculation).
    f) the resistance of the load on the output side

12. A step-down transformer is used to convert $120\ \text{V}$ from the wall source to an audio receiver voltage. If there is a $0.80\ \text{A}$ current on the primary side and the turns ratio is 13:1, find

a) the voltage across the secondary side.
b) the current delivered to the stereo.
c) the resistance of the stereo components.
d) the power delivered to the secondary side.
e) the power delivered to the primary side.

13. A transformer has $100\ \text{turns}$ on its primary side and $600\ \text{turns}$ on its secondary side. It is used to power an elevator motor that requires $2\ \text{A}$ at $6.0 \times 10^2\ \text{V}$. What is the potential difference and current on the primary side of the transformer?

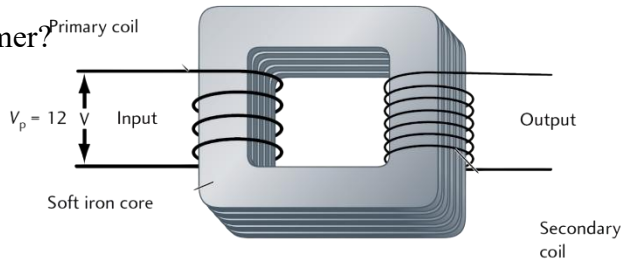
14. A transformer is intended to plug into a standard $120\ \text{V}$ household outlet to provide power for a $6\ \text{V}$ cassette player.

15. a) If the primary coil has $1100\ \text{turns}$, how many turns are required on the secondary side of the transformer?

b) If this was a multipurpose transformer that could be switched to another secondary voltage of $3\ \text{V}$ to power a CD player, how many secondary turns would it require then?

16. A primary voltage of $12\ \text{V}$ AC was applied to the transformer illustrated in **Fig. 18.32.**

a) What is the ratio of primary to secondary turns N_P/N_S?
b) What is secondary voltage?
c) Is this a step-up or a step-down transformer?



17. A neon lamp requires a secondary output voltage of $1.0 \times 10^3\ \text{V}$ at a resistance of $300\ \Omega$. The transformer will run from a standard 120 V outlet.
a) What is the power consumption of the lamp's transformer?
b) What is the current drawn from the primary circuit?
c) What is the N_P/N_S ratio for the transformer coils?

18. Canadian Tire sells a device that operates on your $12\ \text{V}$ car battery and converts the secondary output to a standard $120\ \text{V}$ AC so that you can operate some low power household items, such as a portable personal stereo consuming $60\ \text{W}$.

a) What kind of transformer is in this device?
b) What must happen to your $12\ \text{V}$ DC electricity before it is transformed to $120\ \text{V}$?
c) What is the turns ratio for this transformer?
d) What is the primary current?

19. A cellular phone battery charger has a transformer with 1150 turns on its primary side and 80 turns on its secondary side. If the charger is intended to be used on a standard 120 V line, what potential difference does the cell phone battery receive?

20. A step-down transformer has 750 turns on its primary side and 12 turns on its secondary side. The voltage across the primary side is 720 V.

a) What is the voltage across the secondary side?

b) The current in the secondary side is 3.6 A. What is the current in the primary side?

c) What power is dissipated in this transformer?

21. A step-up transformer has 500 turns on the primary side and $15\ 000$ turns on the secondary side.

a) The potential difference in the secondary side is 3600 V. What is the potential difference on the primary side?

b) The current in the secondary side is 3.0 A. What is the current in the primary side?

c) What power is dissipated in this transformer?

22. To use North American electrical appliances in Europe, you need to take along a transformer to "adapt" to the European 240 V standard voltage (compared to our 120 V standard).

a) To use your 10 A iron in the hotel room, what turn ratio must there be in the transformer?

b) What current will you draw from this European circuit?

c) What would happen if you plugged your iron directly into this circuit without a transformer? Explain.

d) 240 V circuits exist in North America for electric clothes dryers and stoves. What safety measure must electricians adapt in order to prevent someone from mixing up these outlets?

23. At one of our CANDU nuclear generating stations, electricity is generated at 20 kV and transmitted at 230 kV.

a) What type of transformer must this be? What is its turn ratio?

b) If the generator can supply only 60.0 A to the primary side of this transformer, what current must be flowing in the secondary side?

24. Electric transmission lines can transmit 180 kW of power with an effective resistance of only about $0.045\ \Omega$. Power of this magnitude is usually distributed at a stepped-up voltage of 1.1 kV (one kilovolt is 1×10^3 V).

a) What is the effective current in the transmission wires?

b) What power loss is expected with these electrical parameters?

c) What percentage of the original power is lost during the transmission?

d) What should be done to the potential difference on the secondary side of this transformer to reduce power loss even further? To prove your case, choose a new value for potential difference and repeat the power loss calculation.

25. At a new hydroelectric plant, students found out that the generators could supply 500 MW (1 MW is a megawatt or 1×10^6 W), but they were only about 89% efficient in transferring the power from the falling water to electricity.

a) What is the maximum power output that the falling water could supply at 100% efficiency?

b) This power output is due to the loss of gravitational potential energy of the water as it falls. How high must this waterfall be if power was generated by 2.0×10^6 kg of water falling every second?

26. A transformer has 400 turns in the primary coil and 10 turns in the secondary coil. The primary voltage is 250 V and the primary current is 2.0 A.

(a) Calculate the
(i) Secondary voltage; **(ii)** Secondary current, assuming the transformer is ideal.
(b) efficiency of transformers. Explain why, and describe **two** features in transformer design that improve efficiency.

27. A battery charger draws electricity from the 240 V mains supply. The charger contains a transformer, which provides an output of 15 V.

(a) There are 6400 turns on the primary coil of the transformer. Calculate the number of turns on the secondary coil.

(b) Assuming that the transformer is 100% efficient, calculate the current flowing in the primary coil if the output current of the transformer is 2.0 A.

28. Figure 18.55 shows part of a power transmission system. Electricity from the power station is transmitted to end users via transmission cables. The power station has a capacity of 200 MW and produces a voltage of 2 kV. The transmission cable is at 400 kV.

The end users receive a voltage of 250 V.

(a) Is transformer A a step-up or a step-down transformer? Explain your answer.
(b) Assuming that transformers A and B are ideal, and that no energy is lost during transmission, calculate the
(i) current flowing through the transmission cables;
(ii) total current supplied to the end users;
(iii) total energy generated by the power station each day.
(c) Why is electricity transmitted at high voltage?

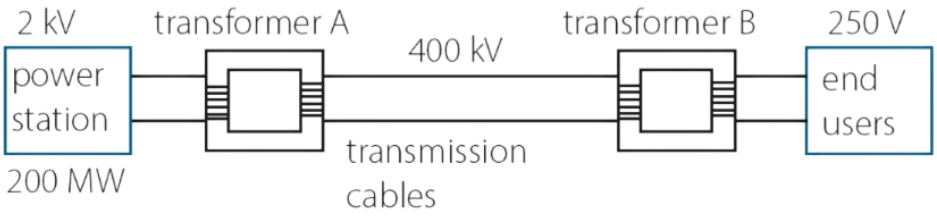


Figure 18.55